ENTERTAINMENT SYSTEM [WITHOUT CENTER DISPLAY]

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Note
- iPod is a registered trademark of Apple Inc.

Outline
- The entertainment system enables listening of radio, CD and music information over the unit equipped on the vehicle.

Function

AM/FM radio
- AM and FM radio broadcasts can be listened to.
- A condenser has been adopted for noise reduction.

RDS (radio data system) (European (L.H.D. U.K.) specs.)
- RDS (radio data system) is available. The RDS (radio data system) displays the selected radio station or the song currently being played at the radio station for the FM radio in the audio unit display.

AUX/USB audio
- By connecting an on-market portable audio USB device such as an iPod to the auxiliary jack,/USB port, audio from the external device can be listened to.

CD playback (With CD player)
- CDs and CD-R (MP3/WMA (Windows Media Audio)/AAC (Advanced Audio Coding)) can be listened to.

Bluetooth® audio (With Bluetooth® unit)
- By pairing a mobile device such as a Bluetooth®-enabled mobile phone or a Smartphone with the audio unit, audio from the Bluetooth®-enabled device can be listened to.

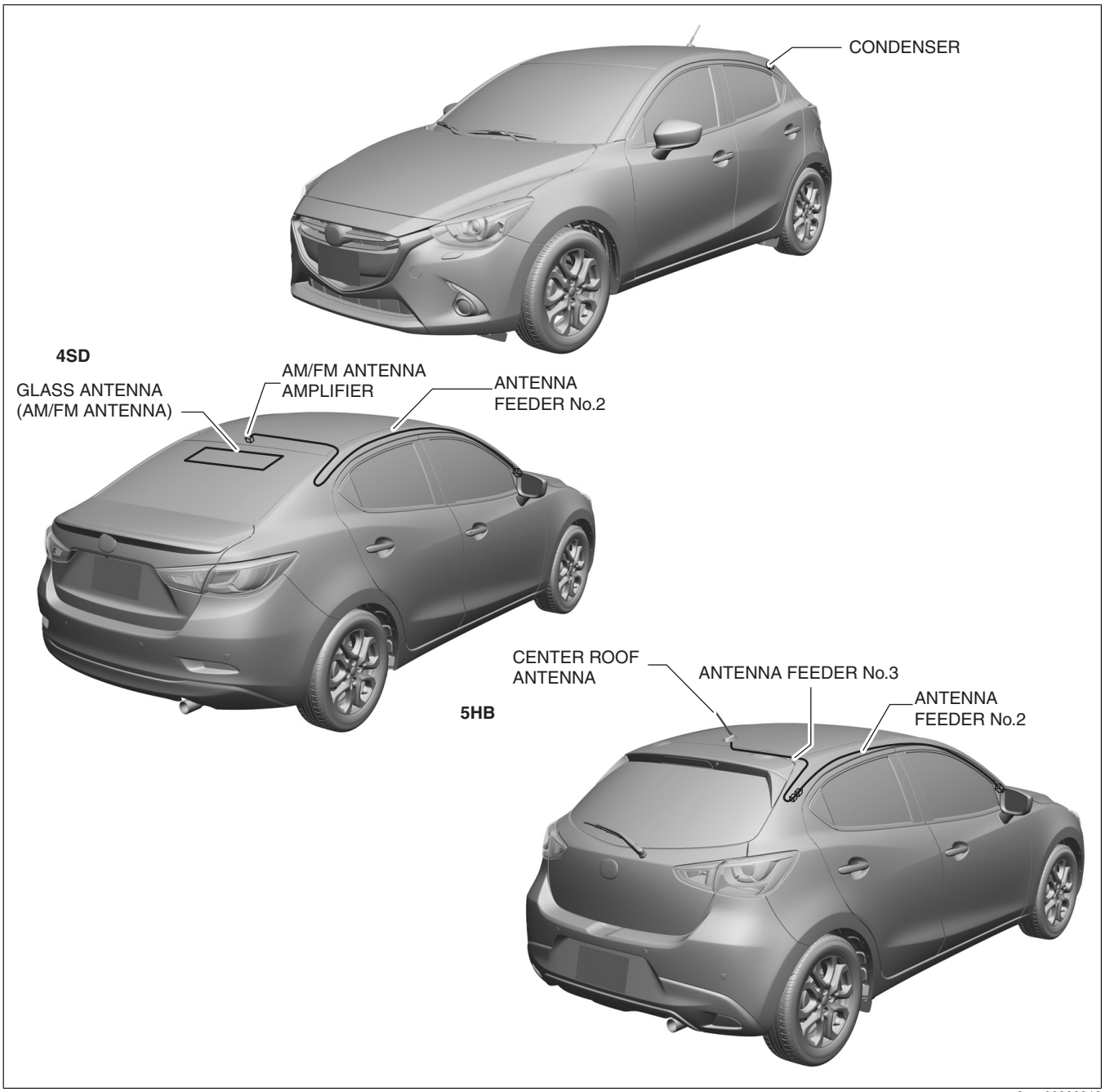
Hands-free telephone (HF/TEL) function (With Bluetooth® unit)
- Answering and receiving telephone calls can be performed without having to directly connect a mobile device such as a Bluetooth®-enabled such as a mobile phone or a Smartphone.

Bluetooth® pairing (With Bluetooth® unit)
- Pairing of a Bluetooth® enabled device can be performed.

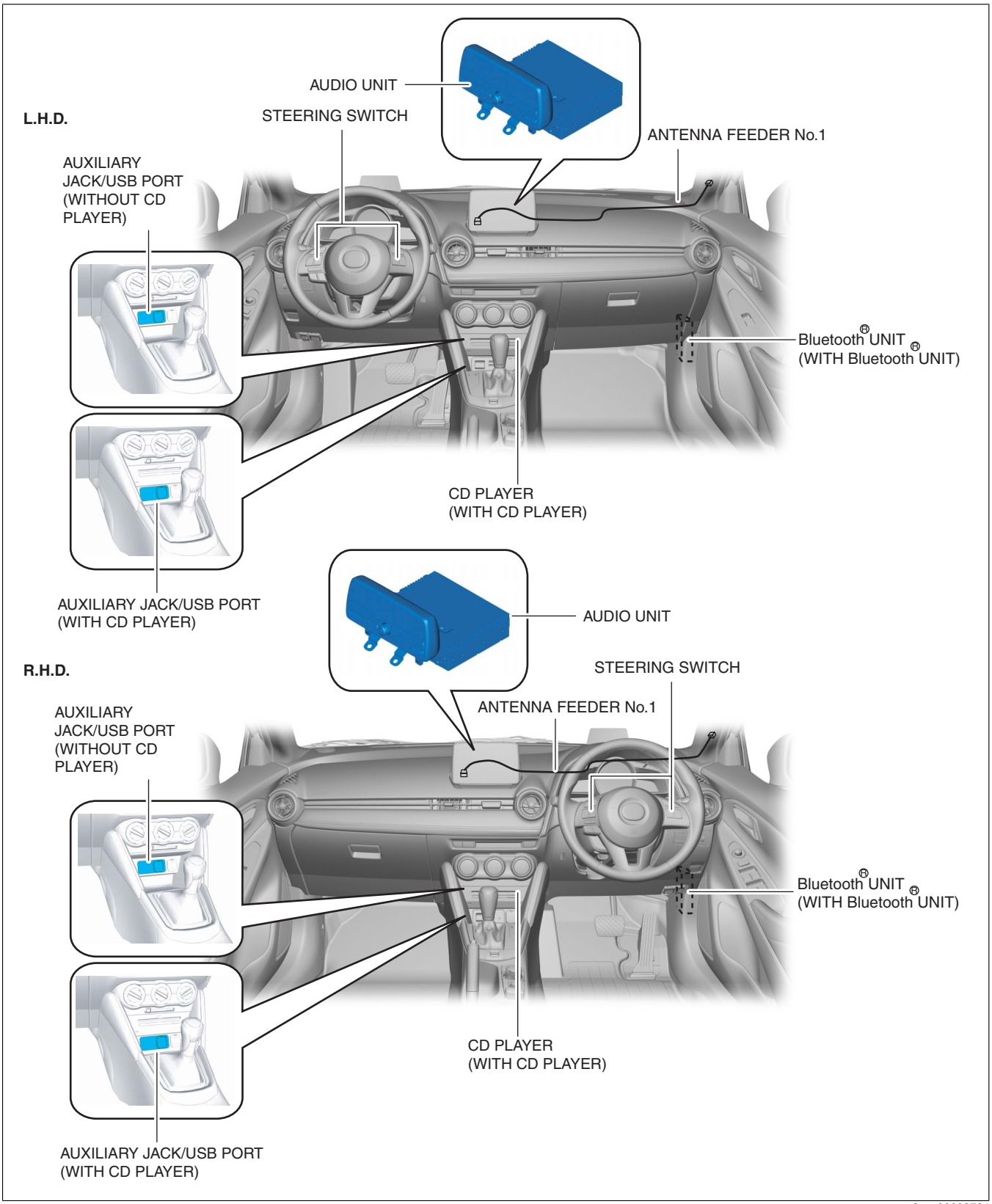
Note
- For the Bluetooth® pairing set up method, Refer to the Workshop Manual for the Bluetooth® PAIRING PROCEDURE.

Voice recognition function (With Bluetooth® unit)
- The Bluetooth® unit is preset with voice commands. If the voice spoken into the microphone for the voice recognition matches the voice command, the Bluetooth® unit performs the task set to the voice command.

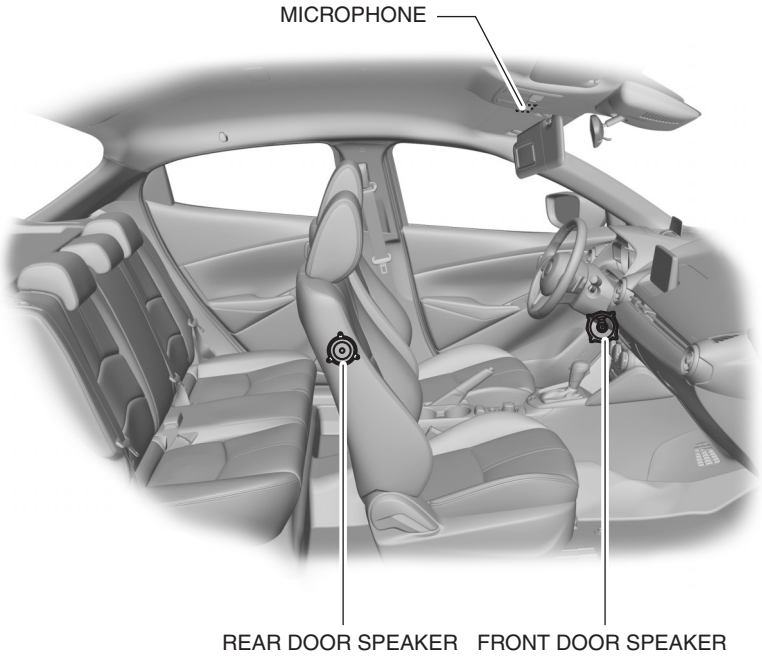
Structural View



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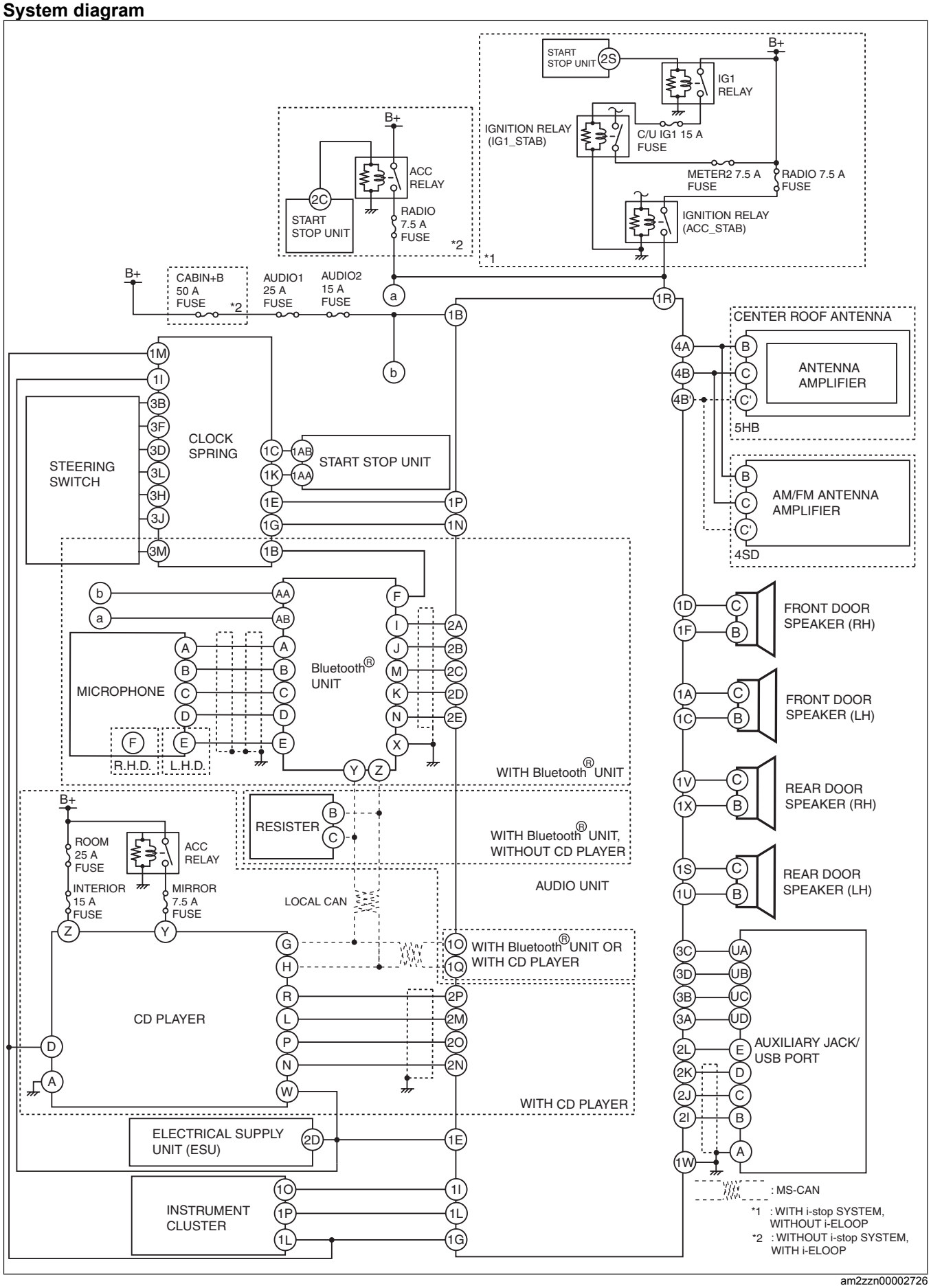


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System diagram

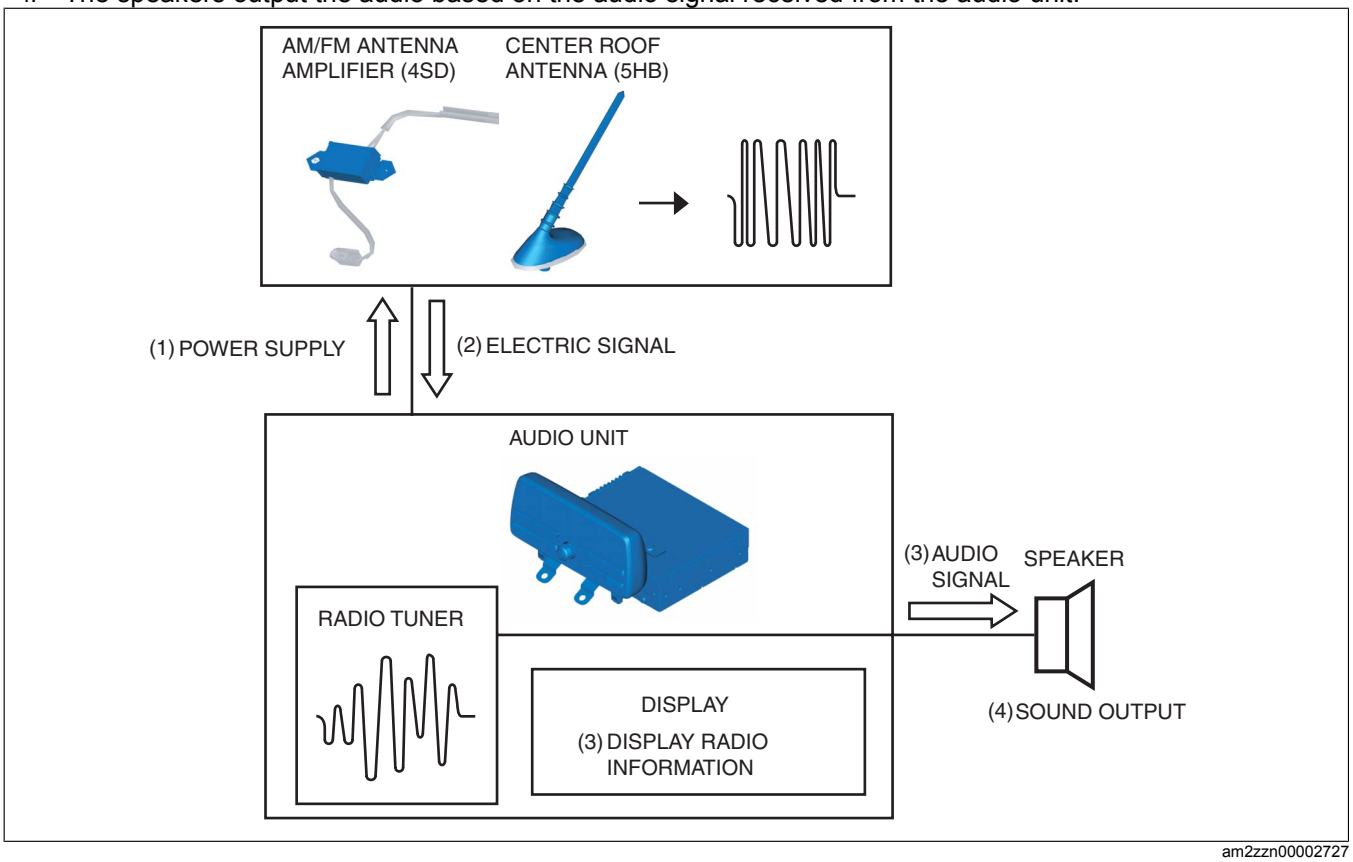


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Operation

AM radio/FM radio/RDS (radio data system) (Except General (R.H.D.))

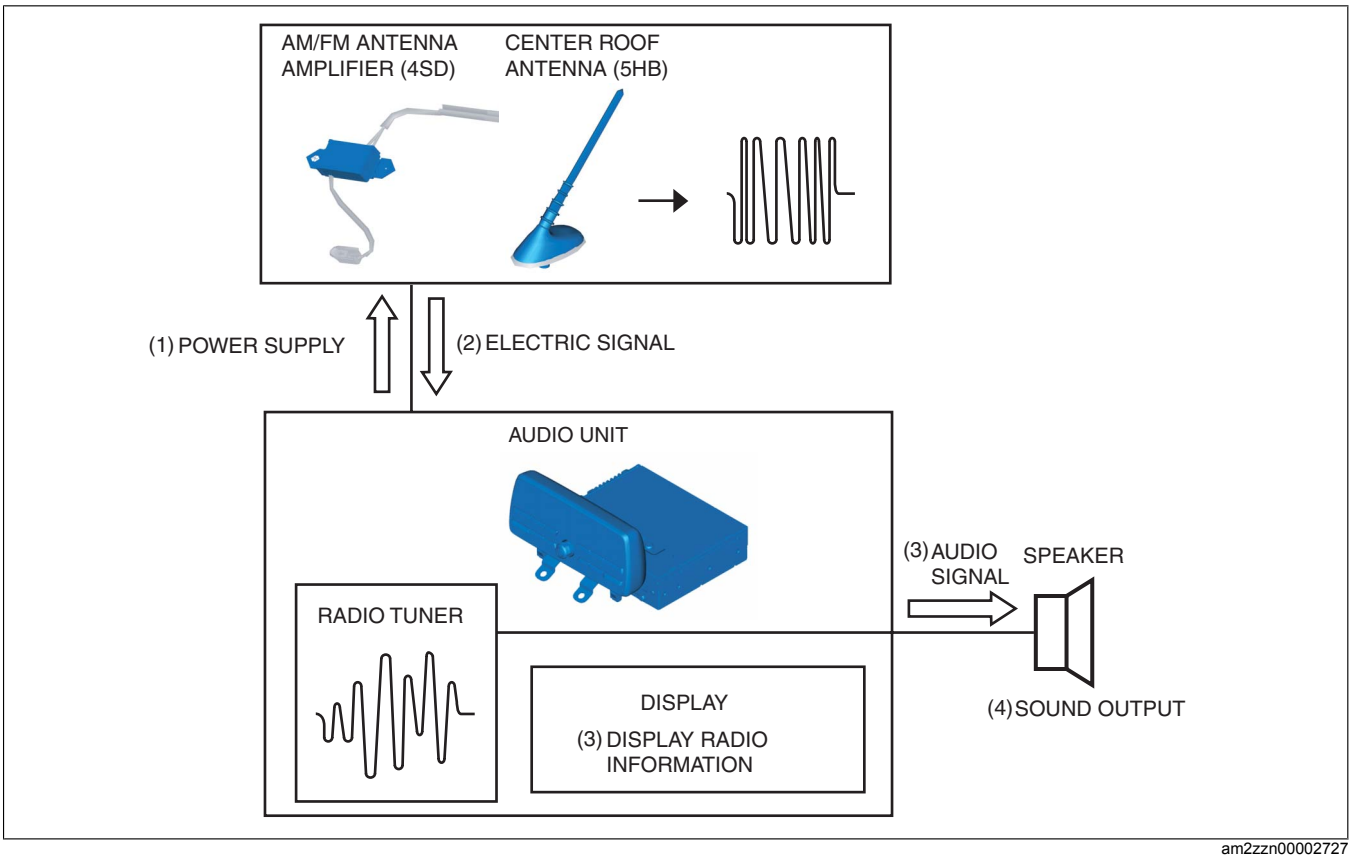
1. When the ignition is switched to ACC or ON (engine off or on), the audio unit supplies power (1) to the AM/FM antenna amplifier (4SD)/center roof antenna (5HB).
2. The audio unit detects (2) the radio broadcast, selected by the user using the radio tuner in the radio unit, from the electrical signal received by the AM/FM antenna amplifier (4SD)/center roof antenna (5HB).
3. The audio unit transmits (3) the audio signal of the detected radio broadcast to the speakers. In addition, the radio information is displayed (3) in the display of the audio unit.
4. The speakers output the audio based on the audio signal received from the audio unit.



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AM radio/FM radio (General (R.H.D.))

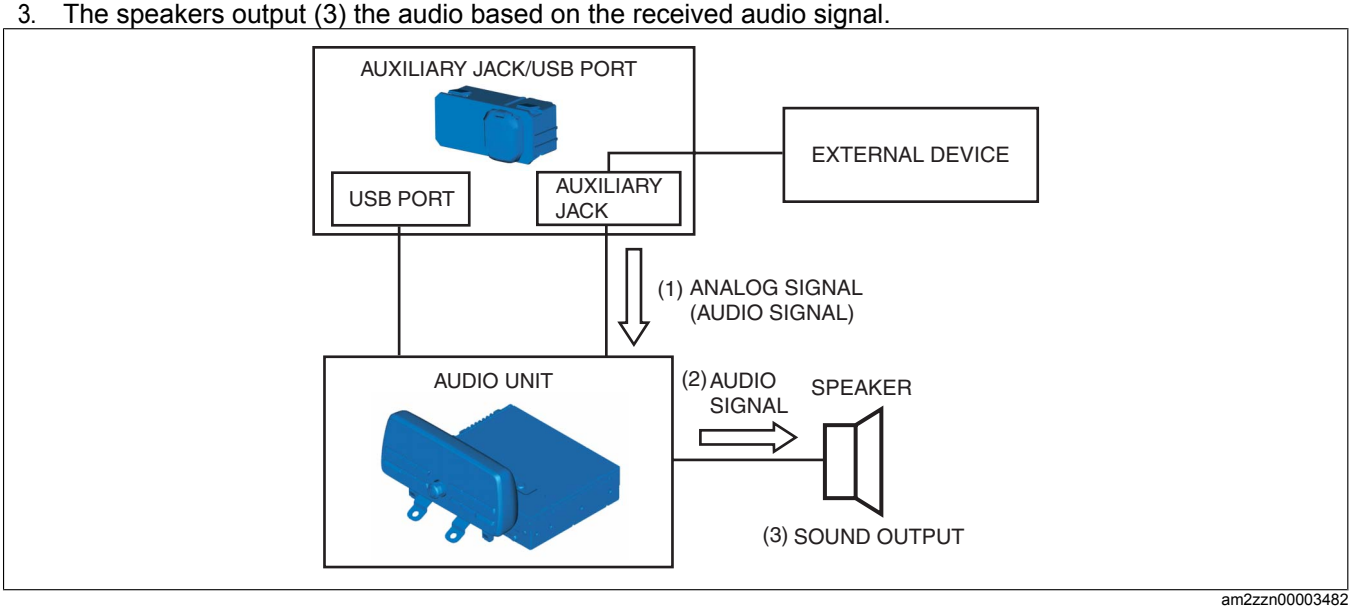
1. If the radio mode is selected when the ignition is switched to ACC or ON (engine off or on), the audio unit supplies (1) power to the AM/FM antenna amplifier (4SD)/center roof antenna (5HB).
2. The audio unit detects (2) the radio broadcast, selected by the user using the radio tuner in the radio unit, from the electrical signal received by the AM/FM antenna amplifier (4SD)/center roof antenna (5HB).
3. The audio unit transmits (3) the audio signal of the detected radio broadcast to the speakers. In addition, the radio information is displayed (3) in the display of the audio unit.
4. The speakers output the audio based on the audio signal received from the audio unit.



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AUX/USB audio (AUX)

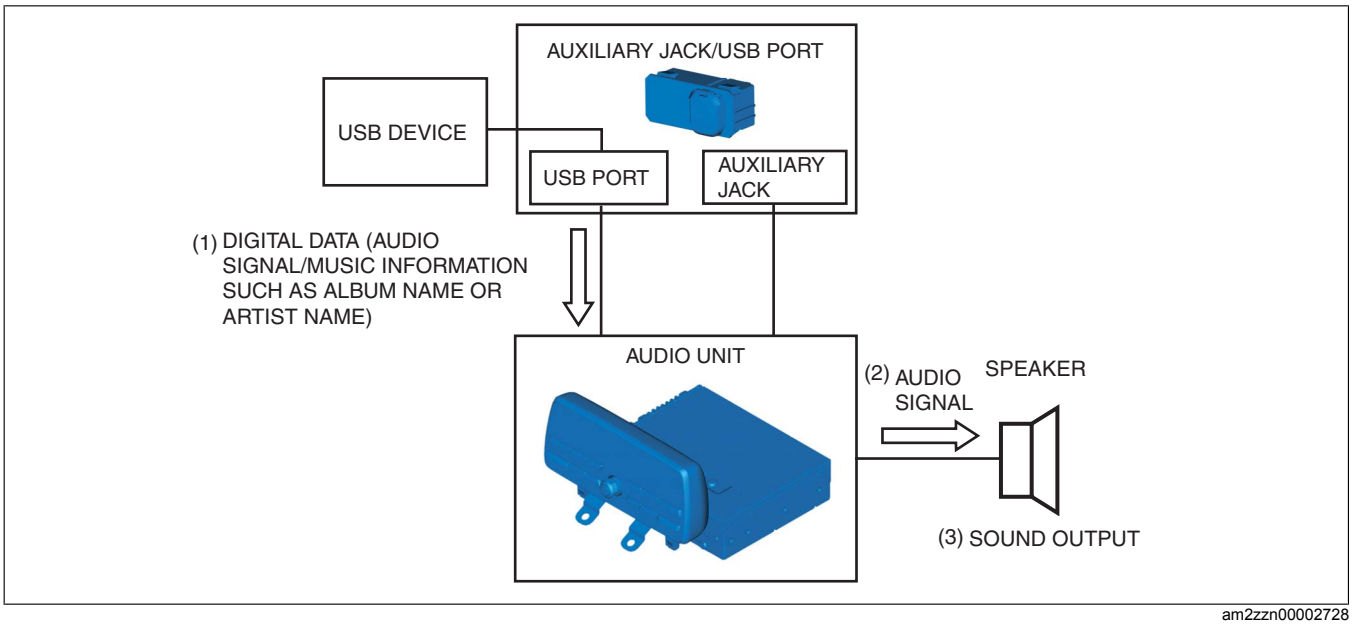
1. When the plug of a portable audio unit is connected to the auxiliary jack/USB port, the auxiliary jack/USB port sends (1) an analog signal (audio signal) from the portable audio device to the audio unit.
2. The audio unit sends (2) the audio signal received from the auxiliary jack/USB port to the speakers.
3. The speakers output (3) the audio based on the received audio signal.



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AUX/USB audio (USB)

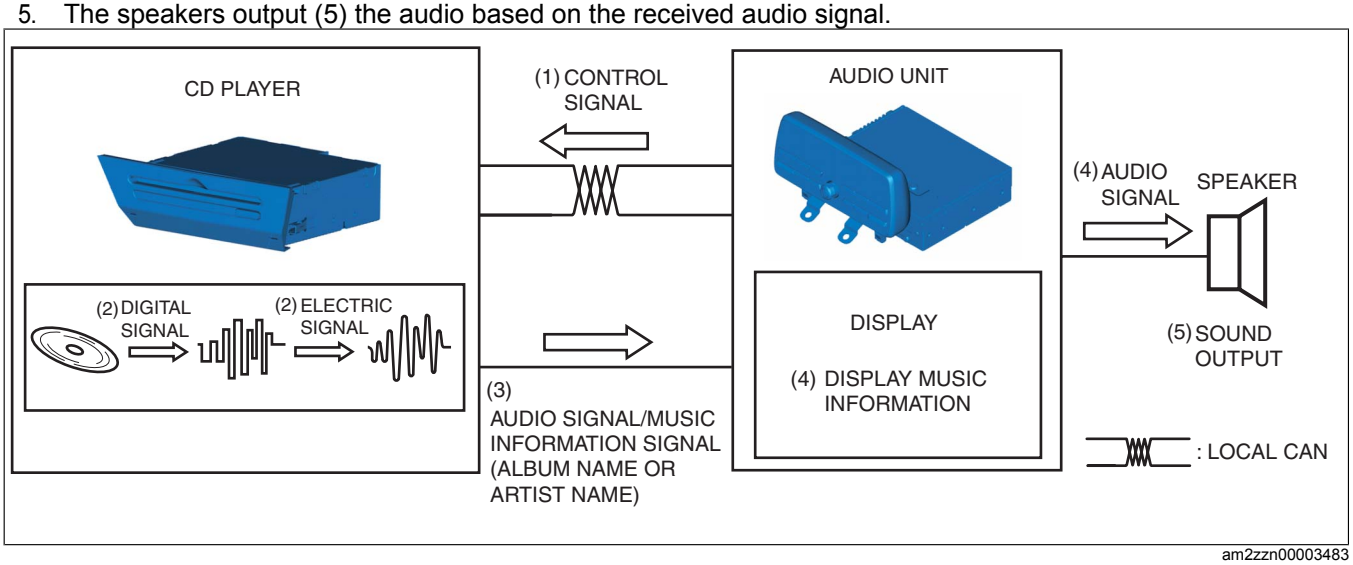
1. When the USB device such as a USB device/iPod is connected to the USB port of the auxiliary jack/USB port, the auxiliary jack/USB port sends (1) the digital data (audio signal/text data signal) of the USB media to the audio unit.
2. The audio unit sends (2) the audio signal received from the auxiliary jack/USB port to the speakers.
3. The speakers output (3) the audio based on the received audio signal.



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CD playback (With CD player)

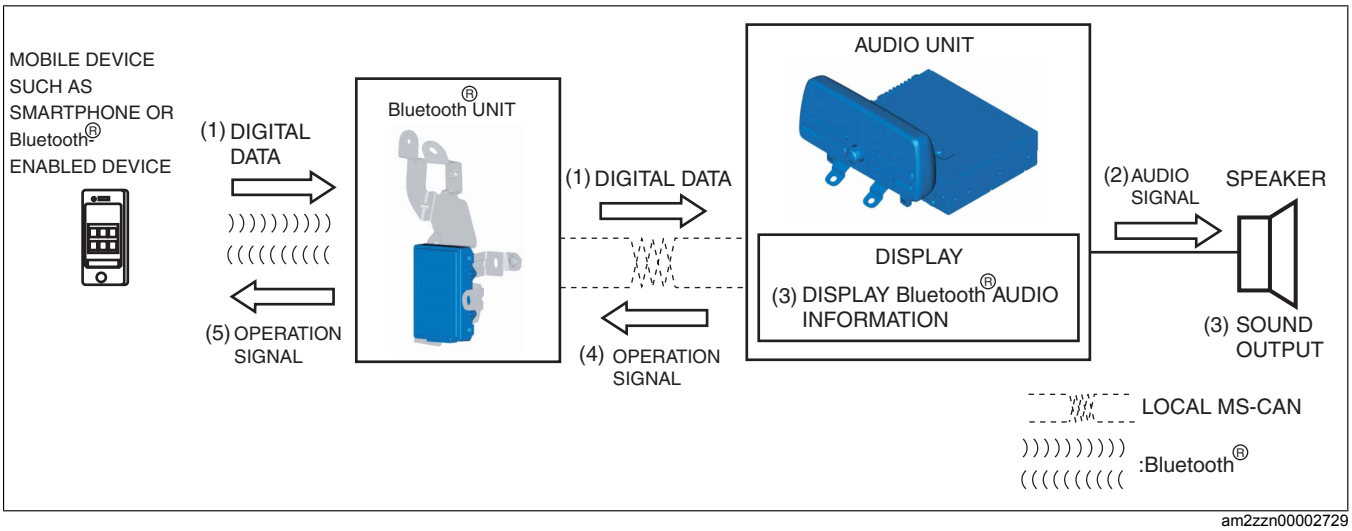
1. When the CD mode is selected, the audio unit sends (1) the control signal to the CD player.
2. The CD player emits laser light to the inserted CD when the control signal is received. The reflected light is detected by the light sensor and the data recorded in the CD is converted (2) from a digital signal to an electric signal.
3. The CD player converts the electric signal to an audio signal and a music information signal, such as album and artist name, using the electrical circuit in the CD player. The converted audio signal and music information signal are sent (3) to the audio unit.
4. The audio unit sends (4) the audio signal received from the CD player to the speakers. In addition, the music information is displayed (4) in the display based on the music information signal
5. The speakers output (5) the audio based on the received audio signal.



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Bluetooth® audio functions (With Bluetooth® unit)

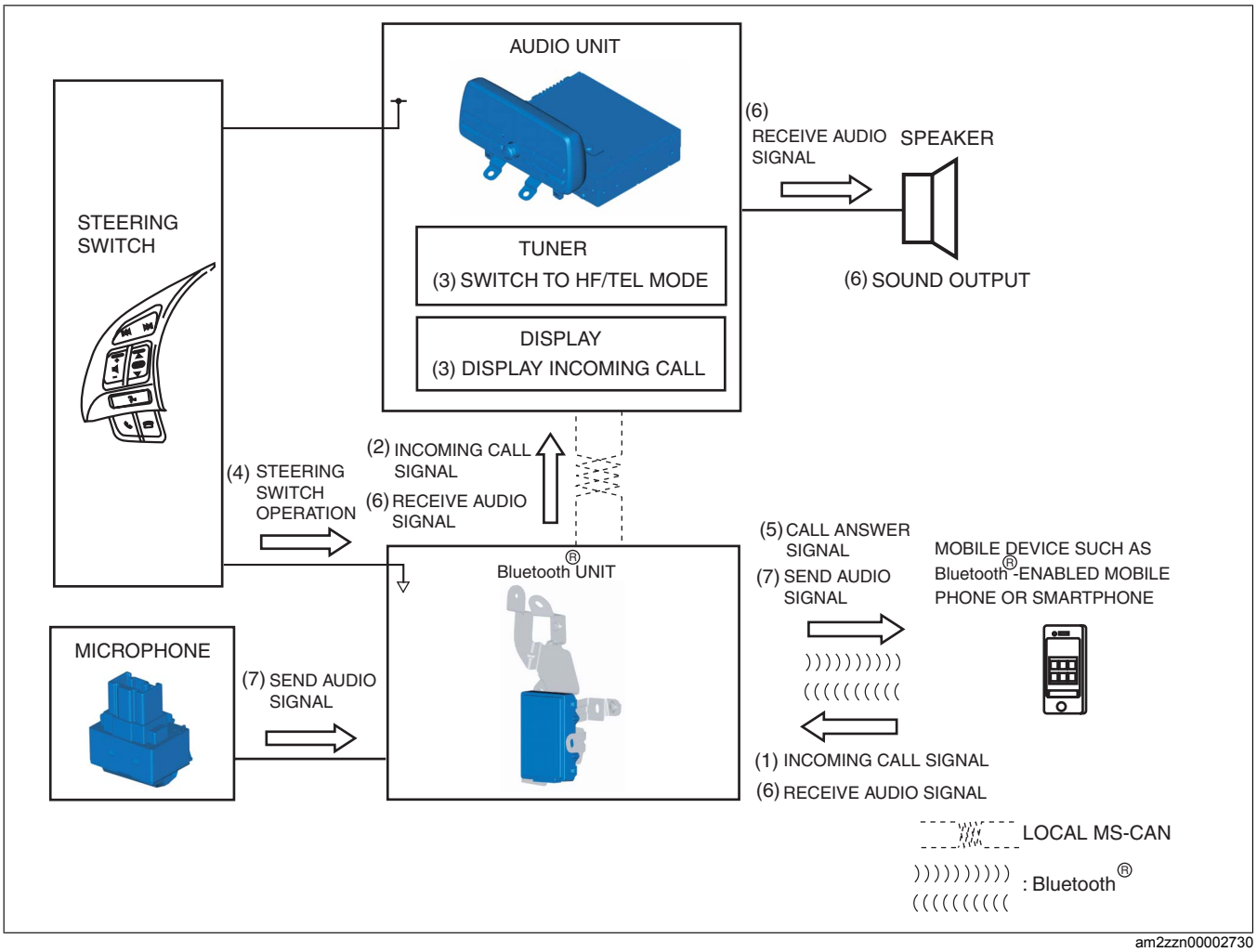
1. The Bluetooth® unit sends (1) digital data received (1) from a mobile device such as a Bluetooth®-enabled mobile phone or a Smartphone to the audio unit.
2. The audio unit sends (2) the audio signal to the audio signal.
3. The audio unit displays (3) the Bluetooth® audio information, and the speakers output (3) the audio based on the received audio signal.
4. When the audio panel on the audio unit is operated, an operation signal is sent (4) to the Bluetooth® unit.
5. The Bluetooth® unit sends (5) the operation signal to the mobile device such as a Bluetooth®-enabled mobile phone or a Smartphone.



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Bluetooth® Hands-free (HF/TEL) function (incoming call) (With Bluetooth® unit)

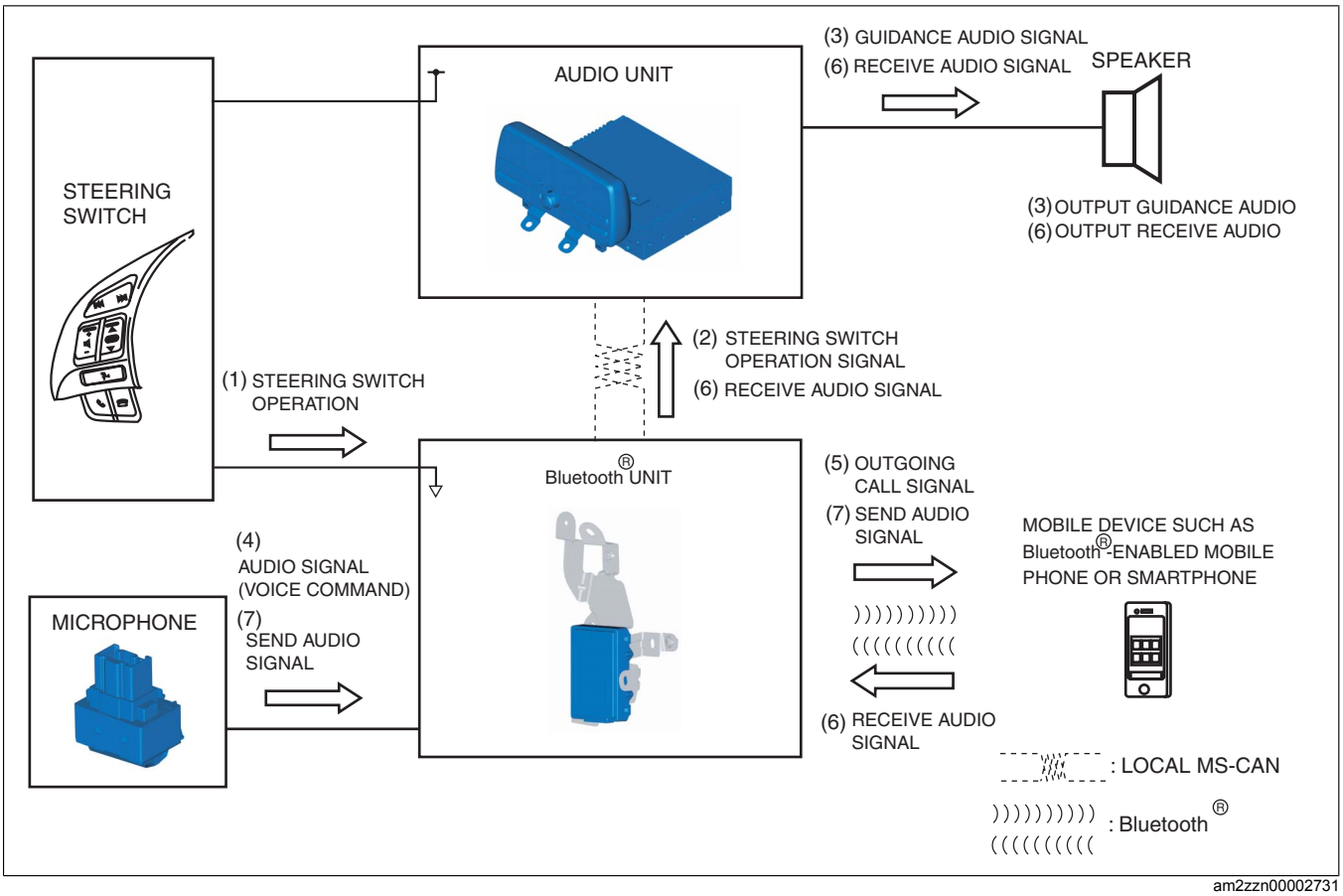
1. When an incoming call is received, a mobile device such as a Bluetooth®-enabled mobile phone or a Smartphone sends (1) the incoming call signal to the Bluetooth® unit.
2. The Bluetooth® unit sends (2) the incoming call signal to the audio unit when it receives the incoming call signal.
3. The audio unit switches (3) the tuner inside the audio unit to HF/TEL mode when it receives the incoming call signal. In addition, the incoming call number is displayed (3) in the display of the audio unit.
4. When the incoming call number is displayed and the TALK button on the steering switch is pressed, the Bluetooth® unit detects (4) the steering switch operation.
5. When the Bluetooth® unit detects the steering switch operation, it sends (5) the call answer signal to a mobile device such as a Bluetooth®-enabled mobile phone or a Smartphone, and the call is initiated.
6. The mobile device, such as a Bluetooth®-enabled mobile phone or a Smartphone, send (6) the received audio signal to the Bluetooth® unit, and the signal is output (6) from the speakers via the audio unit.
7. The voice recognition-use microphone sends (7) the voice signal of the user to the Bluetooth® unit, and the Bluetooth® unit sends (7) the voice signal to a mobile device such as a Bluetooth®-enabled mobile phone or a Smartphone.



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Bluetooth® Hands-free (HF/TEL) functions (outgoing) (With Bluetooth® unit)

1. When the TALK button on the steering switch is pressed, the Bluetooth® unit detects (1) the steering switch operation.
2. When the steering switch operation is detected, the Bluetooth® unit sends (2) a steering switch operation signal to the audio unit.
3. The audio unit sends (3) the guidance audio signal to the speakers and the speakers output (3) the guidance audio.
4. The voice recognition microphone converts the words (voice commands) produced by the user after the guidance audio output to an audio signal and sends it to the Bluetooth® unit.
5. When the Bluetooth® unit receives the audio signal from the voice recognition microphone, it sends (5) an outgoing call signal to a mobile device such as a Bluetooth®-enabled mobile phone or a Smartphone, and the call is initiated.
6. The mobile device, such as a Bluetooth®-enabled mobile phone or a Smartphone, sends (6) the received audio signal to the Bluetooth® unit, and the signal is output (6) from the speakers via the audio unit.
7. The voice recognition microphone sends the voice signal of the user to the Bluetooth®-unit, and the Bluetooth® unit sends (7) the received voice signal to a mobile device such as a Bluetooth®-enabled mobile phone or a Smartphone.



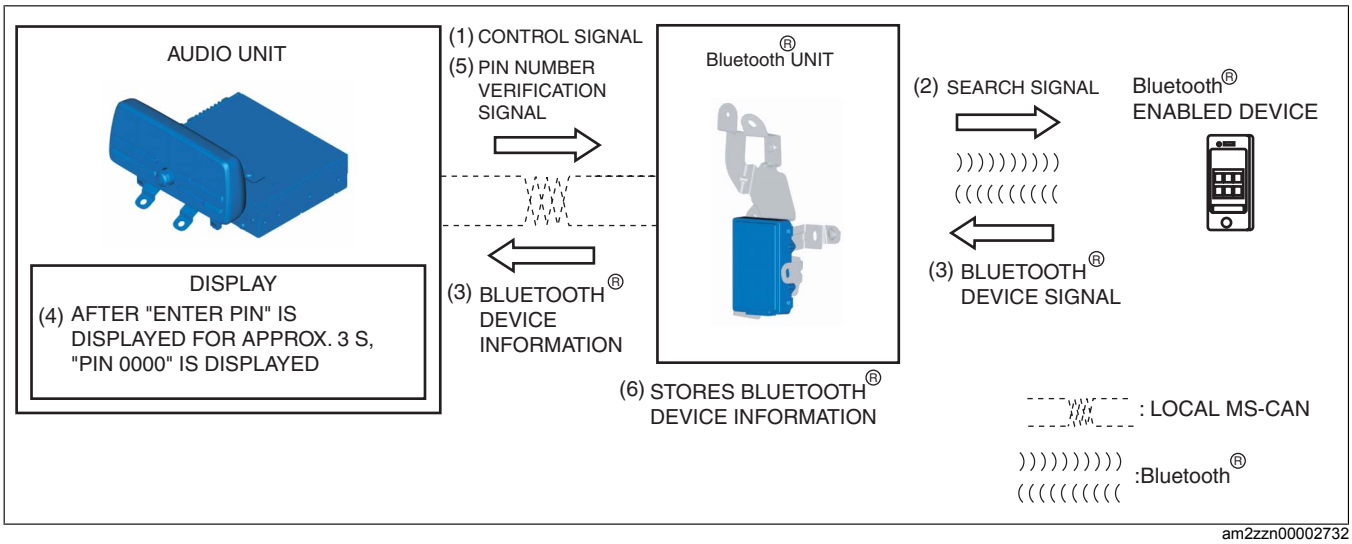
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Pairing with Bluetooth® (With Bluetooth® unit)

1. When the panel button on the audio unit is operated and PAIR DEVICE is selected, the audio unit sends (1) a control signal to the Bluetooth® unit.
2. When the Bluetooth® unit receives the control signal, it sends (2) a search signal to a Bluetooth®-enabled device near the Bluetooth® unit.
3. The Bluetooth®-enabled device near the Bluetooth® unit sends (3) the Bluetooth® device signal to the Bluetooth® unit, and the Bluetooth® unit sends the Bluetooth® device information to the audio unit.
4. When the Bluetooth® device information is received, the audio unit displays (4) “PIN0000” after “ENTERPIN” is displayed for **approx. 3 s** in the display.
5. When the panel button on the audio unit is operated and the PIN number is input, the PIN number verification signal is sent (5) to the Bluetooth® unit.
6. When the PIN number verification signal is received, the Bluetooth® unit stores (6) the Bluetooth®-enabled device information, and the pairing is completed.

Note

- When the pairing is completed, “Musical note symbol” is displayed in the display of the audio unit. In addition, “PAIRSUCCESS” is displayed for **approx. 3 s** and a beep sound is output for **approx. 0.5 s**.
- If 7 Bluetooth® devices (maximum) have been programmed, “MEMORYFULL” is displayed in the audio unit and Bluetooth® pairing is canceled.
- If a Bluetooth® pairing is not done correctly, the display returns to the [PAIRDEVICE] screen after “Err” is displayed for **approx. 3 s**.



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